

Spain

National Commission for Markets and Competition

I. Short summary

This contribution was made by the Spanish National Markets and Competition Commission (CNMC) to the 2nd Annual Report of the Stanford Computational Antitrust addresses the topic of Computational tools for collusion detection.

The CNMC has among its top priorities adapting its competition enforcement to the current wave of digitization and artificial intelligence. This commitment has led to initiatives on two different grounds.

On the one hand, an active analysis and monitoring of digital business models and sectors, including the potential of algorithms and other information technologies to enable collusion and anticompetitive conduct.

On the other hand, the creation of an Economic Intelligence Unit (EIU). Among other things, this unit has the objective of applying new techniques of data analysis artificial intelligence to detect not only new forms of (algorithmic) anticompetitive conduct but also traditional ones. The unit also aims at increasing ex-officio detection and at strengthening the robustness of other investigations.

The CNMC remains committed to progressing in these two fronts in order to ensure that competition is safeguarded in digital activities and throughout the economy. However, the scope of this paper is mainly set on the use of computational tools (including, but not limited to, AI-based tools) by the EIU, which will be summarily described, considering their current stage of development.

II. Introduction

Algorithms and artificial intelligence are widely used by companies to conduct business, often with welfare enhancing effects. However, they create several challenges for competition, and hence for competition authorities.

Algorithms and other information technology tools can be used for anticompetitive purposes, by means of software tools with a varied degree of sophistication (pricing algorithms, etc.). But anticompetitive conduct can also appear

in other forms, such as self-preferencing or discrimination (*e.g.*, in rankings or matching algorithms). These risks appear more often in digital business models where algorithmic tools are used more frequently, but they can also take place in other sectors.

In order to tackle these challenges, the Spanish National Markets and Competition Commission (CNMC) has introduced digitization as a priority in its Strategic Plan,⁴² leading to different initiatives in its annual Action Plans.⁴³ One of such initiatives is the creation of an Economic Intelligence Unit (EIU), to take advantage of new digital techniques as an aid to detect different kinds of anticompetitive conduct, including when this conduct is facilitated by algorithms and other advanced tools.

This contribution summarizes the work carried out by the Economic Intelligence Unit (EIU) in order to increase detection of anticompetitive conduct through digital tools.

III. Detection and analysis of anticompetitive conduct in the Economic Intelligence Unit

The CNMC's priorities, set in its Strategic and Action Plans, include enhancing detection capabilities and the analysis of anticompetitive conducts, among other things by refining the analysis of algorithms and artificial intelligence to gain a better understanding of their potential use in anticompetitive conduct detection.

This is one of the objectives of the Economic Intelligence Unit (EIU) of the CNMC, created in 2018. Given its tasks, the team of the EIU consists of a multidisciplinary group with economic, legal, statistical, mathematical, IT and data protection profiles, all of them with extensive experience in competition matters.

The analysis carried out by the EIU is used both for purely *ex-officio* detection and for refining the analysis in other cases: enhancing and reinforcing evidence, sampling and testing of allegations, dawn-raid planification, etc. Collecting the right data allows them to use specific techniques and analyses, such as machine-learning,

⁴² CNMC, Plan estratégico, <https://www.cnmc.es/sobre-la-cnmc/plan-estrategico> (last accessed June 7, 2023).

⁴³ CNMC, Plan de Actuación, <https://www.cnmc.es/sobre-la-cnmc/plan-de-actuacion> (last accessed June 7, 2023).

graph analysis and other advanced techniques related to Big Data management and processing.⁴⁴

One set of computational aids at EIU's disposal are searching tools based on an extensive data collection activity. Prior to the creation of the EIU itself, the CNMC has been working since 2015 on a public procurement database, creating specific algorithms for data extraction and cleaning, categorizing the data by quality levels regarding all participating bids and bidders.

At this point, we have to notice that, in Spain, public procurement data are stored in different repositories.

Firstly, the Public Sector Contracts Register includes basic information on the contracts awarded by all contracting bodies (central, regional, autonomous bodies, other entities governed by public law) including later amendments.

In addition to it, Public Sector Procurement Platform (PSPP) is a completely up-to-date electronic platform publishing all the calls for tenders and their outcomes. Even if according to law, all central contracting bodies are obliged to use this platform to manage their purchasing processes, regional and local contracting bodies can choose to create their own ones, provided that the platforms of the different government authorities and public entities are interconnected to establish a single platform that centralizes the publication of public sector procurement.

Then, Ministry of Finance Centralized Procurement Portal publishes calls for tenders in processes using framework agreements, centralized contracts and other centralized procedures. Likewise, this is a compulsory method for State-wide contracting bodies but optional for regional and local ones.

Finally, autonomous regions and local entities have established their own procurement data platforms.

With data coming from all these sources, EIU has hence created a complete digital database of public procurement in Spain that has already allowed the development of

⁴⁴ See the CNMC's contribution to the competition policy roundtable on Data Screening Tools for Competition Investigations in December 2022 to learn more about the data use and screens by the EIU. OECD, *Data Screening Tools for Competition*, <https://www.oecd.org/daf/competition/data-screening-tools-for-competition-investigations.htm> (last accessed June 7, 2023).

several proprietary, powerful search tools that take advantage of the processed data and currently help EIU to detect potential cases of bid-rigging.

Quite recently, the access to non-awarding data (losing bids) has opened the way to the use of machine-learning techniques that will make possible to decisively promote *ex officio* detection.

One of the working fields is the exploration and development and training of an AI-tool based on machine-learning algorithms, both supervised and semi-supervised, to determine the best set of features (both factual and statistically calculated) that will help the machine to build a practical model in an attempt to classify the behavior of companies participating in public procurement processes, eventually leading to the detection of collusion schemes and their classification by their specific nature or pattern, and, up to a point, the *ex-ante* prediction of such anticompetitive conditions on some given situations.

The second line of work in this field will be the design and building of graph databases, that store nodes and relationships instead of tables, or documents. Such databases will help in the task of analyzing relations among activity sectors, companies, and the company leaders themselves. That would hopefully provide EIU with some clustering capabilities, aiming to the prior identification of groups of companies, or even individuals with the potential of becoming a real cartel, or any other sort of collusion scheme. Both action lines are currently under intensive development, not being fully operational yet.

Other investigative approach at the EIU, different from the others but which takes advantage of some of the techniques mentioned above, is the use of Open-Source Intelligence (OSINT) and Human Intelligence (HUMINT) tools in order to provide an accurate identification and location of organizations and persons of interest, the relationships among them, and their degree of control of the companies and organizations which are under close scrutiny.

This work by the EIU is key to improve detection and investigation, and it should increase *ex-officio* detection of cartels, and any other anticompetitive conduct. In this regard, it is worth mentioning that between 2018 and 2022 70% of CNMC's cartel decisions have not been triggered by a leniency application. This increased rate of *ex-officio* cartel investigations since 2018 is partly due to the special scrutiny of bid-rigging cases (it is also important to be noted that 70% of CNMC's cartel cases concern bid-rigging).

IV. Conclusion

The enforcement of competition law in digital markets is complex, especially as much as algorithms and artificial intelligence are concerned. Competition policy tools are flexible enough to adapt to the disruption driven by digitization, but some challenges remain.

The CNMC has tried to adapt and refine its enforcement tools, for instance, through the creation of an Economic Intelligence Unit. The CNMC has also given strategic priority to monitoring anticompetitive conduct driven by algorithms and digital business models. There is a real need for advanced computational tools based on similar techniques, to ensure a more even ‘playing field’ among cartelists and competition authorities, and therefore some of the current efforts of CNMC are set on that target.